

Practical 2

Data Wrangling II

Create an “Academic performance” dataset of students and perform the following operations using Python. 1. Scan all variables for missing values and inconsistencies. If there are missing values and/or inconsistencies, use any of the suitable techniques to deal with them.

2. Scan all numeric variables for outliers. If there are outliers, use any of the suitable techniques to deal with them.

3. Apply data transformations on at least one of the variables. The purpose of this transformation should be one of the following reasons: to change the scale for better understanding of the variable, to convert a non-linear relation into a linear one, or to decrease the skewness and

```
import pandas as pd
import numpy as np
from pathlib import Path
```

```
[ ]: df = pd.read_csv('/content/xAPI-Edu-Data.csv')
df
```

```
[ ]:
gender NationalTy PlaceofBirth StageID GradeID SectionID \
0      M      KW      KuwaIT  lowerlevel  G-04      A
1      M      KW      KuwaIT  lowerlevel  G-04      A
2      M      KW      KuwaIT  lowerlevel  G-04      A
3      M      KW      KuwaIT  lowerlevel  G-04      A
4      M      KW      KuwaIT  lowerlevel  G-04      A
..      ...      ...      ...      ...      ...
475    F      Jordan      Jordan  MiddleSchool  G-08      A
476    F      Jordan      Jordan  MiddleSchool  G-08      A
477    F      Jordan      Jordan  MiddleSchool  G-08      A
478    F      Jordan      Jordan  MiddleSchool  G-08      A
479    F      Jordan      Jordan  MiddleSchool  G-08      A
```

```
Topic Semester Relation  raisedhands  VisITedResources \
0      IT      F  Father      15      16
1      IT      F  Father      20      20
2      IT      F  Father      10      7
3      IT      F  Father      30      25
4      IT      F  Father      40      50
..      ...      ...      ...      ...
475  Chemistry      S  Father      5      4
476  Geology      F  Father      50      77
```

477	Geology	S	Father	55	74
478	History	F	Father	30	17
479	History	S	Father	35	14

	AnnouncementsView	Discussion	ParentAnsweringSurvey	\
0	2	20		Yes
1	3	25		Yes
2	0	30		No
3	5	35		No
4	12	50		No
--	...	...	...	
475	5	8		No
476	14	28		No
477	25	29		No
478	14	57		No
479	23	62		No

	ParentschoolSatisfaction	StudentAbsenceDays	Class
0	Good	Under-7	M
1	Good	Under-7	M
2	Bad	Above-7	L
3	Bad	Above-7	L
4	Bad	Above-7	M
--	...	...	...
475	Bad	Above-7	L
476	Bad	Under-7	M
477	Bad	Under-7	M
478	Bad	Above-7	L
479	Bad	Above-7	L

[480 rows x 17 columns]

```
[ ]: # 1. SCAN & DEAL WITH MISSING VALUES AND INCONSISTENCIES
print("--- Step 1: Missing Values ---")
print(df.isnull().sum())    # Scan for nulls

# Handle Inconsistencies: standardizing categorical text (strip spaces & fix_
s_casing)
for col in df.select_dtypes(include='object').columns:
    df[col] = df[col].str.strip().str.title()
print("\nInconsistencies handled (Text standardized to Title Case).")
```

```
--- Step 1: Missing Values ---
gender          0
NationalITy     0
PlaceofBirth    0
StageID         0
```

```

GradeID          0
SectionID        0
Topic            0
Semester         0
Relation         0
raisedhands      0
VisITedResources 0
AnnouncementsView 0
Discussion       0
ParentAnsweringSurvey 0
ParentschoolSatisfaction 0
StudentAbsenceDays 0
Class           0
dtype: int64

```

Inconsistencies handled (Text standardized to Title Case).

```

[ ]: # 2. SCAN & DEAL WITH OUTLIERS
print("\n--- Step 2: Outliers (IQR Method) ---")
numeric_cols = df.select_dtypes(include=[np.number]).columns

for col in numeric_cols:
    Q1, Q3 = df[col].quantile(0.25), df[col].quantile(0.75)
    IQR = Q3 - Q1
    lower, upper = Q1 - 1.5 * IQR, Q3 + 1.5 * IQR

    # Identify count
    outlier_count = ((df[col] < lower) | (df[col] > upper)).sum()
    print(f"{col}: {outlier_count} outliers found.")

    # Handle: Capping (Winsorization)
    df[col] = np.clip(df[col], lower, upper)

```

```

--- Step 2: Outliers (IQR Method) ---
raisedhands: 0 outliers found.
VisITedResources: 0 outliers found.
AnnouncementsView: 0 outliers found.
Discussion: 0 outliers found.

```

```

[ ]: # 3. DATA TRANSFORMATION
# Applying Log Transformation to 'raisedhands' to decrease skewness
df['raisedhands_log'] = np.log1p(df['raisedhands'])

print("\n--- Step 3: Transformation (Log) ---")
print(df[['raisedhands', 'raisedhands_log']].head())

```

--- Step 3: Transformation (Log) ---

	raisedhands	raisedhands_log
0	15	2.772589
1	20	3.044522
2	10	2.397895
3	30	3.433987
4	40	3.713572

[]: